UNIT-V

Lasers and Fibre Optics

Introduction

Laser is a light source which amplifies intensity of light to produce highly directional, coherent and monochromatic beam of light. Lasers are now days used in different fields like medical, industrial, communication and military etc. The acronym (or) short form for LASER is "Light amplification by Stimulated Emission of Radiation" and laser is a specialized light source which is different from conventional light such as tube light (or) electric bulb.

5.1 Characteristics of Lasers:

Lasers when compared with conventional light, it possesses a few outstanding characteristics.

They are

(1) Laser is highly monochromatic

(2) Laser is highly directional

(3) Laser is highly coherent

(4) The intensity of laser is very high

Monochromaticity: The light emitted by a laser is more monochromatic than that of any conventional monochromatic source. The monochromaticity of a light source is measured by its degree of non-monochromaticity. For a good laser the degree of non-monochromaticity is of the order of 10^{-13}.where as the degree of non-monochromaticity for a conventional monochromatic light source is of the order 10^{-3}.Therefore by comparing these two values, it is clear that laser is highly monochromatic.

Directionality: Laser emits light in a single direction. The directionality of the laser beam is measured in terms of divergence, hence for getting high directionality there should be low divergence.

Coherence: The light rays emitting from the laser are in phase with each other, therefore the laser is highly coherent.

Intensity: Due to the concentration of energy over a small region laser beam becomes more intense. The intensity of a laser beam is measured in terms of number of photons emitted per unit area per sec.

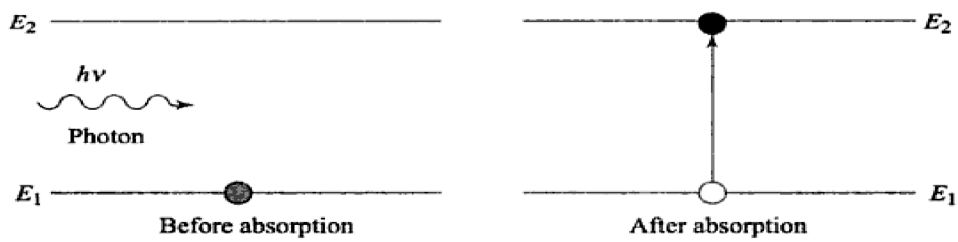
The number of photons emitted by a good laser is in the range of 10^{22} to 10^{34} photons per m^2-sec.But the no.of photons emitted by any other light source is in the range of 10^{16} photons/ m^2-sec.So by comparision it is clear that laser is highly intense.

5.2 Interaction of Radiation with matter

Before knowing about spontaneous emission we need to know some fundamental concepts which are explained below:

Absorption:

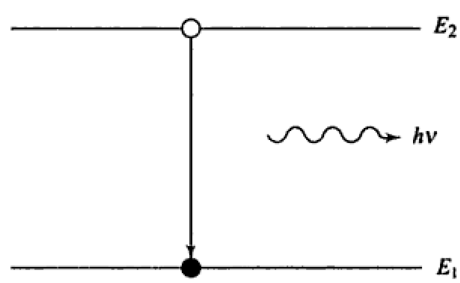
The process of absorption of energy when a particle transfers from it's ground state to higher energy state is called as absorption.



Life time:

The duration of time spent by an atom in the excited state is known as life time of that energy state.

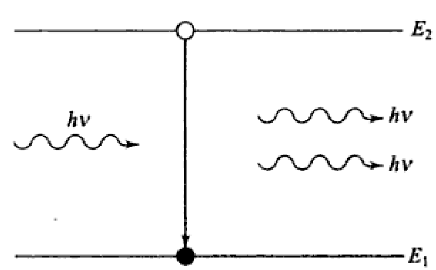
Spontaneous Emission:



The emission of light photon after the lifetime without any inducement during the transition of atoms from higher energy level to lower energy level is called as spontaneous emission.

Here frequency of emitted photon, $\upsilon = \dfrac{E_2 - E_1}{h}$

The photons in this case have various wavelengths and they are out of phase. Therefore the photons are incoherent.



Stimulated Emission:

Emission of light photon by the inducement of a photon having energy equal to the emitted photon's energy or the energy difference between the transmission energy levels is called as stimulated emission.

In stimulated emission, the two emitted photons will have same energy and phase. So therefore they are coherent and they have single wavelength.

Metastable State:

The excited state which has long lifetime is known as metastable state.

5.3 Einstein's Coefficients or Einstein's relation

Einstein coefficients explain about the transmissions of electrons among the energy levels in terms of probability.

Absorption:

If we consider a two level energy system, the rate of probability to occur absorption process from energy level 1to energy level 2 depends on properties of energy level 1 and 2 and is proportional to incident energy density U(v) of the radiation of frequency υ incident on the atom.

$$\therefore \quad P_{12} \alpha U(v)$$

$$\Rightarrow \quad P_{12} = B_{12} U(v) \text{ ---------------------- >(1)}$$

Where B_{12} is proportionality constant represents properties of energy levels known as Einstien's coefficient of absorption.

Spontaneous Emission:

The rate of probability to occur spontaneous emission process from energy level 2 to energy level 1 depends only on the properties of energy levels 1 and 2.This process is independent of energy density $U(v)$.

$$\therefore \quad (P_{21})_{spontaneous} = A_{21} \text{ -------------------- >(2)}$$

Where A_{21} is proportionality constant, represents properties of energy levels known as Einstein's coefficient of spontaneous emission.

Stimulated emission:

The rate of probability to occur stimulated emission process from energy level 2 to energy level 1 depends on properties of energy levels 1 and 2 as well as proportional to stimulated energy density $U(v)$ of frequency ϑ incident on the atom.

$$\therefore \quad (P_{21})_{stimulated} \propto U(v)$$

$$(P_{21})_{stimulated} = B_{12}U(v) \text{ ------------------- >(3)}$$

Where B_{12} is proportionality constant, represents properties of energy levels known as Einstien's coefficient of Stimulated emission.

The total transition probability of atoms from energy level 2 to energy level 1 can be written as

$$P_{21} = (P_{21})_{spontaneous} + (P_{21})_{stimulated}$$

$$P_{21} = A_{21} + B_{12}U(v) \text{ ---------------- >(4)}$$

Relation between Einstein Coefficients:

Let us consider N_1 and N_2 be populations in the energy level 1 and 2 respectively in a system of atoms, which is at thermal equilibrium at a temperature T.

The no. of atoms that take transitions per unit volume from energy level 1 to energy level 2 in unit time can be written as

$$N_1P_{12} = N_1B_{12}U(v) \text{ ----------------------- >(5)}$$

The no. of atoms that take transitions per unit volume from energy level 2 to energy level 1 in unit time can be written a

$$N_2P_{21} = N_2[A_{21} + B_{12}U(v)] \text{ -------------- >(6)}$$

At equilibrium, the no. of transitions from energy level 1 to energy level 2 will be equal to the no. of transitions from energy level 2 to energy level 1.

$$\therefore \quad N_1P_{12} = N_2P_{21} \text{ ------------------- >(7)}$$

From eq^{n}s (5) & (6)

We have,

$$N_1B_{12}U(v) = N_2[A_{21} + B_{12}U(v)]$$

$$\Rightarrow \quad N_1B_{12}U(v) - N_2\, B_{12}U(v) = N_2A_{21}$$

$$\Rightarrow [N_1B_{12} - N_2\, B_{12}]B_{12}U(v) = N_2A_{21}$$

$$\Rightarrow \quad U(v) = \frac{N_2A_{21}}{N_1B_{12} - N_2\, B_{12}}$$

$$\Rightarrow U(v) = \frac{N_2A_{21}}{B_{21}\left[\frac{N_1}{N_2}\left(\frac{B_{12}}{B_{21}}\right) - 1\right]}$$

$$\Rightarrow \qquad U(v) = \frac{A_{21}}{B_{21}} \cdot \frac{1}{\left[\frac{N_1}{N_2}\left(\frac{B_{12}}{B_{21}}\right) - 1\right]} \text{ -------------- >(8)}$$

But we know that, according to Boltzmann's distribution law

$$N_1 = N_0 exp\left[\frac{-E_1}{k_B T}\right] \text{ ---------------- >(10)}$$

Where N_0 **is population in the ground state and** k_B is Boltzmann's constant

$$\frac{N_1}{N_2} = exp\left[\frac{E_2 - E_1}{k_B T}\right]$$

$$\frac{N_1}{N_2} = exp\left[\frac{h\vartheta}{k_B T}\right] \quad (\because E_12 - E_11 = h\vartheta) \text{ ---------------- >(11)}$$

Substitute eq^n (11) in eq^n (8)

$$U(v) = \frac{A_{21}}{B_{21}} \frac{.1}{\left\{exp\left[\frac{h\vartheta}{k_B T}\right]\left(\frac{B_{12}}{B_{21}}\right) - 1\right\}} \text{ ------------------- >(12)}$$

But according to Planck's radiation law,

$$U(v) = \frac{8\pi h\vartheta^3}{C^3} \cdot \frac{1}{\left\{exp\left[\frac{h\vartheta}{k_B T}\right] - 1\right\}} \text{ --------------- >(13)}$$

By comparing eq^n.s (12) & (13)

$$\left.\begin{array}{ll} \frac{A_{21}}{B_{21}} = \frac{8\pi h\vartheta^3}{C^3} & \frac{B_{12}}{B_{21}} = 1 \\ \frac{A_{21}}{B_{21}} \propto \vartheta^3 & B_{12} = B_{21} \end{array}\right\} \text{ --------------- >(14)}$$

Eq^n (14) shows the relations between Einstein's coefficients B_{12}, B_{21} and A_{21}.

The first relation shows that the ratio of Einstein's coefficients of A_{21} and B_{21} is proportional to cube of the frequency of incident photon. The second relation shows the rate of probability of induced emission and absorption are equal when the system is in equilibrium.

5.4 Population inversion:

The no of atoms per unit volume in an energy level is known as population of that energy level.

If N is the no of atoms per unit volume in an energy state E then the expression for population can be written as

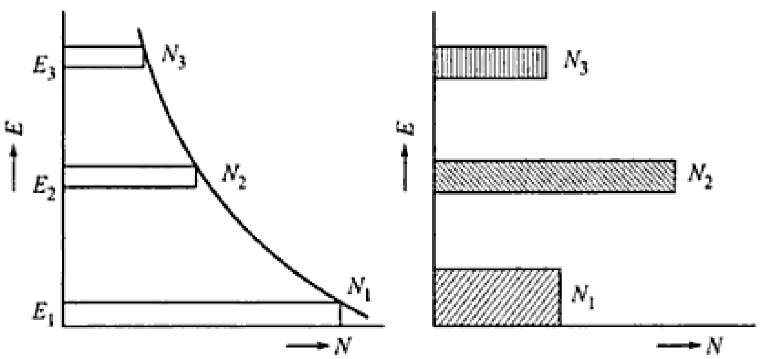
$$N = N_0 exp\left[\frac{E}{k_0 B}\right]$$

From the equation of population, population is maximum in the ground state and decreases exponentially as energy level increases.

So population inversion is a process in which the population of a particular higher energy state is made more populated than that of a lower energy state.

In order to understand the concept of population inversion consider three level system in which there are three energy levels E_1, E_2 and E_3 ,population in those energy levels are N_1, N_2 and N_3 respectively.In normal conditions $E_1 < E_2 < E_3$ and $N_1 > N_2 > N_3$.

We know that E_1 is the ground state and its lifetime is unlimited and E_3 is highest energy state and it is most unstable state and its lifetime is very less. Whereas E_2 is an excited state and has more lifetime compared to E_3.Therefore here E_2 is metastable state.When some energy is supplied to the system, the atoms excite from ground state (E_1) to excited states (E_2 &E_3) .Due to unstability, excited atoms will come back to ground state after the lifetime of the respective energy states E_2 and E_3.If this process is continued then atoms will excite continuously to E_2 and E_3.Because E_3 is the most unstable state,atoms will fall into E_2 immediately.At a particular point,the population in E_2 will become more than the population in ground state.



(a) Normal population of a system ($N_1 > N_2$)

(b) Inverted population of a system ($N_2 > N_1$)

Population of atoms

5.5 Block Diagram of a Laser System

The block diagram of a laser system contains three components. They are
(1)Source of energy
(2)Active medium
(3)Optical cavity

Source of Energy:

It supplies energy to the active medium to achieve population inversion i.e., it performs pumping process.

Active medium:

It is a place where the metastable state is achieved. In metastable state only the population inversion takes place. It can be a liquid, solid, gas or PN-junction.

Optical Cavity:

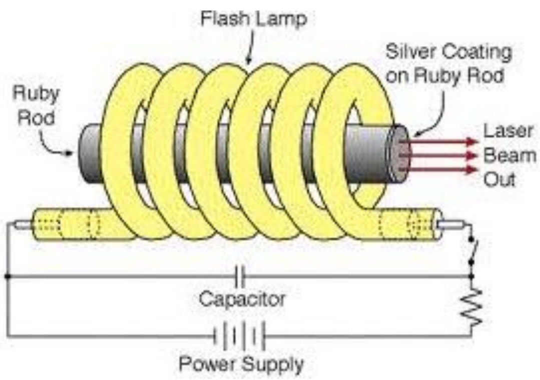
It is an enclosure of active medium and consists of two mirrors. Here one mirror is fully reflective and the other one is partially transparent. Due to mirrors arrangement emitted laser takes back and forth reflections until it gains sufficient energy to come out. The output laser beam is emitted from partially transparent mirror.

5.6 Laser systems

5.6.1 Ruby Laser

Ruby Laser is constructed by Maiman in 1960.It is a pulsed laser. The duration of each pulse is about 10 nanoseconds.

Construction: Ruby laser is made up of a cylindrical ruby crystal rod of composition Al_2O_3 which is doped with 0.05% of Cr_2O_3.The ends of the Ruby rod are silvered such that one end is fully reflecting and the other end is partially reflecting. A Xenon flash tube is arranged around the Ruby rod as shown in the below figure, which supplies flash light of wave length 5600A^0 to the active medium to achieve population inversion.But here only a part of the flash light is used for pumping of chromium atoms while the rest heats up the apparatus. Therefore a cooling arrangement is provided to keep the experimental set up at normal temperature.



Therefore here,

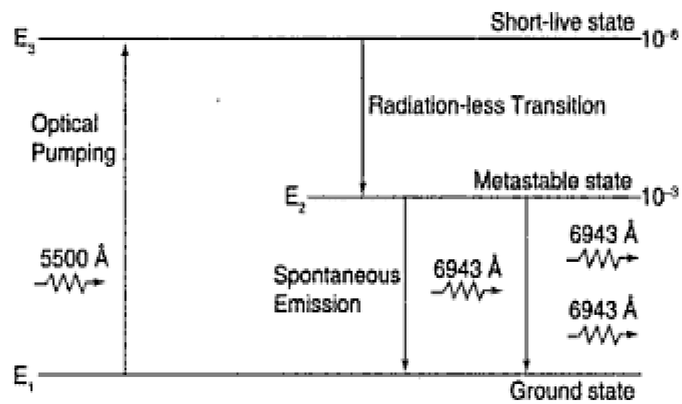
Source of Energy → Xenon flash light

Active medium → Ruby Crystal rod

Optical Cavity → Arrangement of Silver polished surfaces on either sides of Ruby rod

Working: In ruby crystal, chromium atoms are responsible for the population inversion. So here chromium atoms have three active energy levels, they are named as Ground state(G), Metastable state(M) and Higher energy state(H).Due to supply of Xenon flash light to Ruby rod, chromium atoms are excited to H energy state.

From higher energy state all of them do not return to the ground state , but some will return to state G and some will move to state M. Now here the transitions from level H to M are non-radiative i.e. the chromium atoms gives part of the energy to crystal lattice in the form of heat. The atoms which are present in this state M have a little chance to go to the ground energy state because it's lifetime is more. Due to continuous process of excitation of atoms to the higher levels, the population in state M will become more, at this stage, the transition occurs from M to G level emitting out photons. Now stimulated emission starts and photons are emitted in different directions. But only those photons will come out of the laser which travel parallel to the axis of the tube while the photons that travel in other directions will pass out of the Ruby. Now this photon beam which is parallel to the axis of the crystal rod grows in strength and comes through the partial reflector and serves as output laser beam. The wavelength of this beam is 6943A^0



Drawbacks:

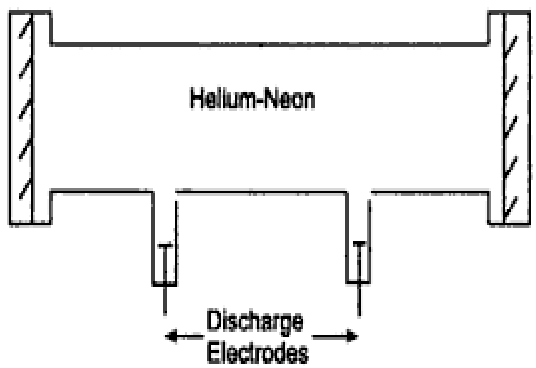
(1)Photons of 6943A^0 are absorbed by the ground state atoms from the laser beam.

(2)The output laser beam is discontinuous.

5.6.2 He-Ne Laser

It is a gas laser and is a four level laser system. Here electric discharge is used as an efficient method for producing population inversion.

Construction: This laser system consists of a gas discharge tube and it is filled with a mixture of Neon (Ne) and Helium (He).The ratio of He-Ne mixture is about 10:1.The gas mixture of Helium and Neon forms the lasing medium and this mixture is enclosed between a set of parallel mirrors forming an optical cavity. One of the mirrors is completely reflecting and the other is partially reflecting in order to amplify the output laser beam.



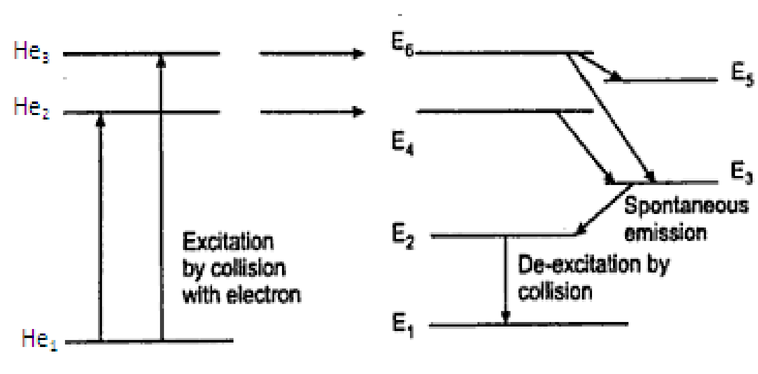
Therefore, here

Source of Energy → Discharge tube

Active medium → He-Ne gas mixture

Optical Cavity → arrangement of reflector

Working: In this laser, the lasing action is due to the Neon atoms. Here Helium atoms are used for pumping of Neon atoms to upper energy levels. When fast moving electrons collide with Helium atoms, then they are excited to the upper states.



Energy Level Diagram.

Here Helium atoms have three active energy levels He_1,He_2& He_3 and Neon atoms have six active energy levels Ne_1-------Ne_6.When He atoms collide with electrons they are excited to upper states He_2 and He_3.These are metastable states. So these atoms which are present in He_2 and He_3 states interact with neon atoms which are in the ground state. The interaction excite the neon atoms to their metastable states (Ne_4 and Ne_6).As the energy exchange continues, the population of neon atoms in the excited states increases more and more. So after reaching metastable state, stimulated emission starts emitting out photons and the excited neon atoms deexcite to the ground state in three different ways. They are

(1)Transition from Ne_6 level to Ne_5 level gives rise to radiation of wavelength 3.39μm.

(2)Transition from Ne_6 level to Ne_3 level gives rise to visible radiation of wavelength 6328A^o this lies in red region.

(3)The transition from Ne_4 level to Ne_3 level gives rise to a wavelength of 1.15 μm ,this again lies in infrared region.

The atoms in Ne_5 and Ne_3 level undergo spontaneous transitions to Ne_2 level, finally Ne atoms comes to ground state through collisions with the walls of the tube. This transition is radiationless. The gas lasers are found to emit light which is more directional and monochromatic. Gas lasers are capable of operation continuously without need of cooling.

5.6.4 Semiconductor laser

A semiconductor diode laser is a specifically made p-n junction diode that emits coherent light under forward bias. R.N. Hall and his coworkers made the first semiconductor laser in 1962. P-N junction lasers are emits light almost anywhere in the spectrum from UV to IR.

The laser diode (light amplification by stimulated emission of radiation) produces a monochromatic (single Color) light. Laser diodes in conjunction with photodiodes are used to retrieve date from-compact discs.

Diode lasers are remarkably small in size (0.1 mm long). They have high efficiency of the order of 40%. In spite of their small size and low power requirements, they produce power outputs equivalent to that of He-Ne lasers. Diode lasers are useful in optical fibre communications, in CD players, CD-ROM drives, optical reading and high speed laser printing etc wide.

Semiconductor Materials

vi) Semiconductors are two different groups, direct band gap semiconductors and indirect band gap semiconductors.

vii) Direct band gap semiconductors are formed by group III_V elements and group IV-VI elements. Most of the compound semiconductors belong to this group.

viii) Leasers are made using direct band gap semiconductors. Gallium Arsenide (GaAs) diode is an example of semiconductor diode laser.

ix) Direct band gap semiconductor is the one in which a conduction band electron can recombine directly with a hole in the valence band. The recombination process lead to emission of light.

Principle

i) The energy hand structure of a semiconductor consists of a valence band and a conduction band separated by an energy gap, E_g. The conduction band contains electrons and the valence band contains holes and electrons.

ii) When an electron from the conduction band jumps into a hole in the valence band, the excess energy E_g is given out in the form of a photon.

iii) Electron-hole recombination is the basic mechanism responsible for emission of light.

Iv) The wavelength of the light is given by the relation $\lambda = hc/E_g$.

x) Semiconductors having a suitable value of E_g emit light in the optical region.

Types of Semiconductor Diode Lasers

Broadly there are two types of semiconductor diode lasers. They are known as *homojunction semiconductor laser and heterojunction semiconductor lasers.*

Homojunction Semiconductor Laser

Heetrojunction Semiconductor Laser

Homojunction Semiconductor Laser

A Simple diode laser which makes use of the same semiconductor material on both sides of the junction is known as a homojunction diode laser. Example: Gallium arsenide (GaAs) laser.

Construction: Fig. shows the schematic of a homojuncation diode laser. Starting with a heavily doped n-type GaAs material, a p-region is formed on its top by diffusing zinc atoms into it. A heavily zinc doped layer constitutes the heavily doped p-region.

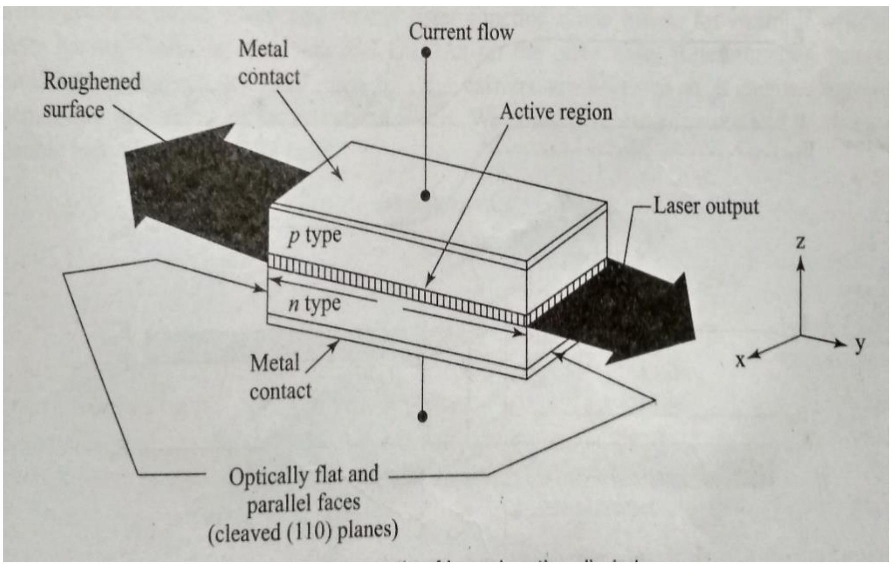


Fig: Homojunction diode laser

The diode is extremely small in size. Typical diode chips are 500 μm long and about 100 μm wide and thick. The top and bottom faces are metalized and metal contacts are provided to pass current through the diode. A pair of parallel planes cleaved at the two ends of the PN junction provides required reflection to from the cavity. The two remaining sides of the diode are roughened to remove lasing action in that direction. The entire structure is packaged in small case which looks like the metal case used for discrete transistors.

Working:

i) Heavily doped p- and n- regions are used. Because of very high doping on n-side, the donor levels are broadened and extend into the conduction band.

ii) The Fermi level is pushed into the conduction band and electrons occupy the levels lying below the Fermi level.

v) Similarly, on the heavily doped p-side the Fermi level lies within the valence band and holes occupy the portion of the valence band that lies above the Fermi Level.

vi) At thermal equilibrium, the Fermi level is uniform across the junction.

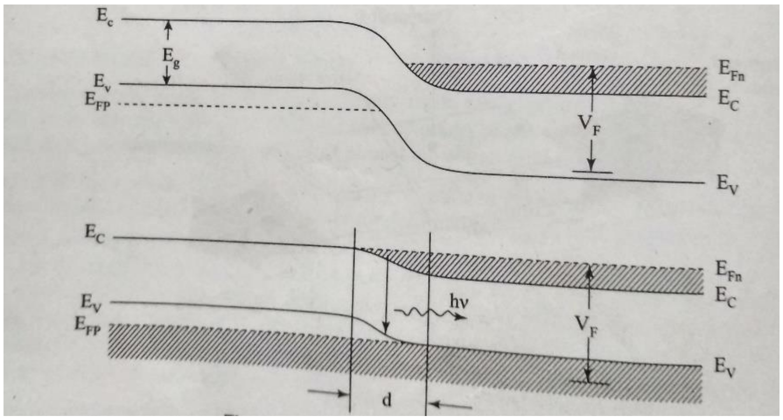


Fig. Laser diode under forward bias

Pumping Mechanism

i) When the junction is forward-biased, electrons and holes are injected into the junction region in high concentrations.

ii) When the diode current reaches a threshold value (see Fig.) , the carrier concentration in the junction region will rise to a very high value.

Population inversion

i) The region d as shown in Fig. contains a large concentration of electrons within the conduction band and simultaneously a large number of holes within the valence band.

ii) The upper energy levels in the narrow region are having a high electron population while the lower energy levels in the same region are vacant. It creates the condition of population inversion and is called an inversion region or active region.

Lasing action

i) Chance recombination leads to emission of spontaneous photons which stimulate the conduction electrons to jump into the vacant states of valence band.

ii) This stimulated electron-hole recombination produces coherent radiation.

iii) GaAs laser emits light at a wavelength of 9000 A in IR region.

Drawbacks of homojunction lasers

i) The active region is not well defined due to the diffusion length of the carriers.

ii) The semiconductor has nearly uniform refractive index throughout. Therefore, light can diffuse from active layer into the surrounding medium. As a result the cavity losses increase.

iii) High threshold currents are required and the laser cannot be operated continuously at room temperature.

5.6.5 CO2 Molecular gas laser

It was the first molecular gas laser developed by Indian born American scientist Prof.C.K.N.Pillai.

It is a four level laser and it operates at 10.6 μm in the far IR region. It is a very efficient laser.

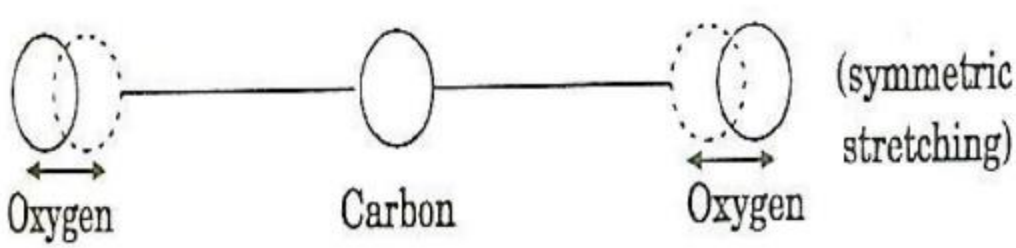
Energy states of CO2 molecules.

A carbon dioxide molecule has a carbon atom at the center with two oxygen atoms attached, one at both sides. Such a molecule exhibits three independent modes of vibrations. They are

a) Symmetric stretching mode.
b) Bending mode
c) Asymmetric stretching mode.

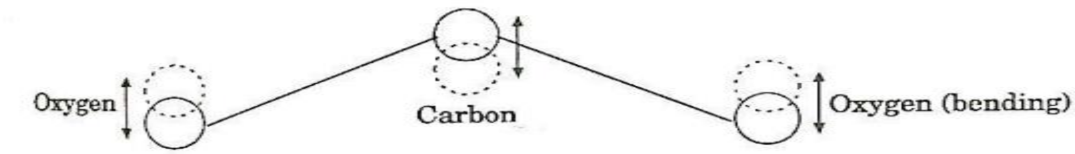
a. **Symmetric stretching mode**

In this mode of vibration, carbon atoms are at rest and both oxygen atoms vibrate simultaneously along the axis of the molecule departing or approaching the fixed carbon atoms.

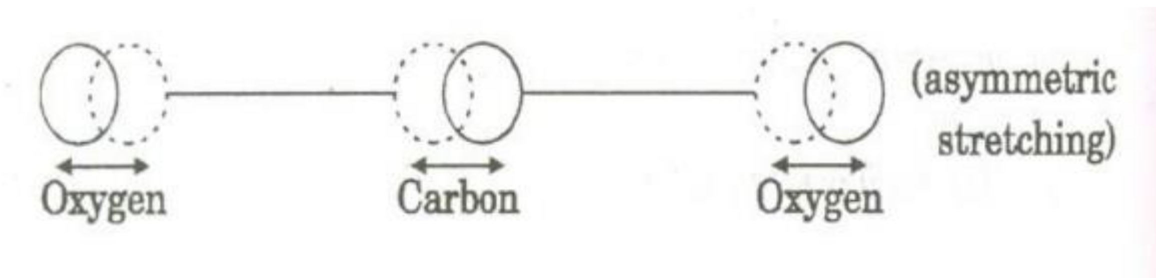


b. **Bending mode:**

In this mode of vibration, oxygen atoms and carbon atoms vibrate perpendicular to molecular axis.



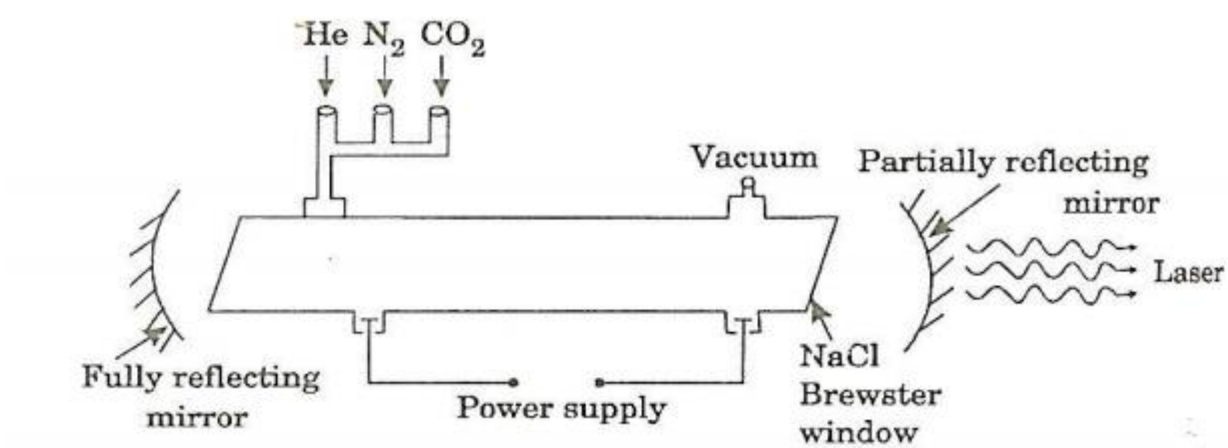
c. **Asymmetric stretching mode**:



In this mode of vibration, oxygen atoms and carbon atoms vibrate asymmetrically, i.e., oxygen atoms move in one direction while carbon atoms in the other direction.

Principle:

The active medium is a gas mixture of CO2, N2 and He. The laser transition takes place between the vibrational states of CO2molecules.



Construction:

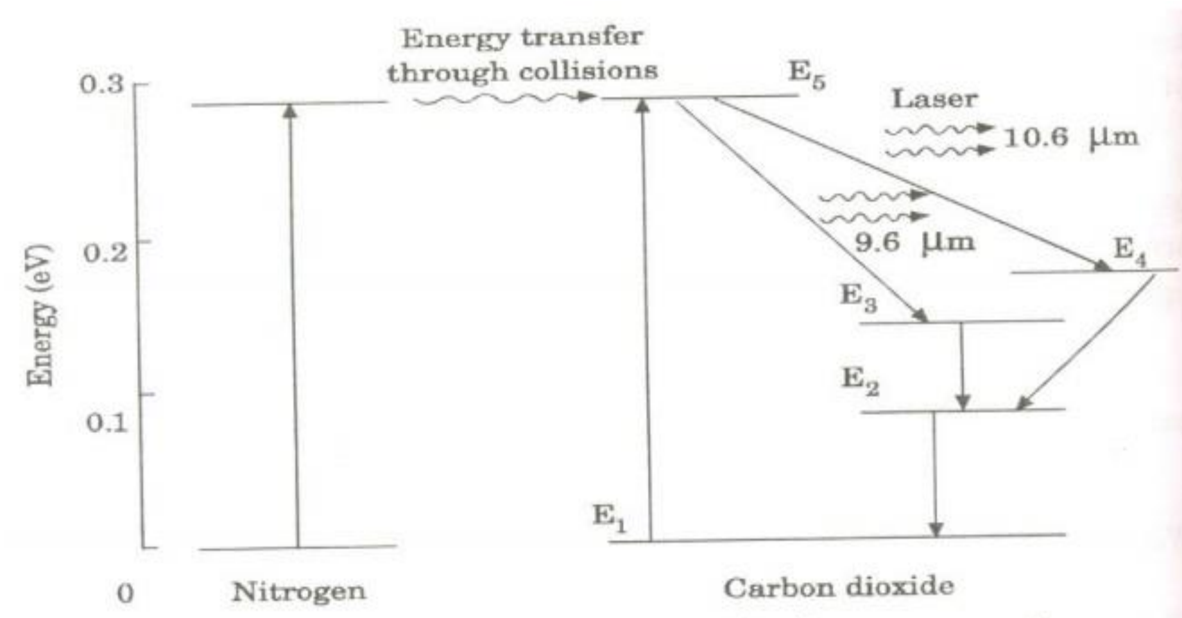
It consists of a quartz tube 5 m long and 2.5 cm in the diameter. This discharge tube is filled with gaseous mixture of CO2(active medium), helium and nitrogen with suitable partial pressures.

The terminals of the discharge tubes are connected to a D.C power supply. The ends of the discharge tube are fitted with NaCl Brewster windows so that the laser light generated will be polarized.

Two concave mirrors one fully reflecting and the other partially form an optical resonator.

Working:

Figure shows energy levels of nitrogen and carbon dioxide molecules.



When an electric discharge occurs in the gas, the electrons collide with nitrogen molecules and they are raised to excited states. This process is represented by the equation

N2 + e* = N2* + e

N2 = Nitrogen molecule in ground state e* = electron with kinetic energy
N2* = nitrogen molecule in excited state e= same electron with lesser energy
Now N2 molecules in the excited state collide with CO2 atoms in ground state and excite to higher electronic, vibrational and rotational levels.
This process is represented by the equation N2* + CO2 = CO2* + N2

N2* = Nitrogen molecule in excited state. CO2 = Carbon dioxide atoms in ground state CO2* = Carbon dioxide atoms in excited state N2 = Nitrogen molecule in ground state.

Since the excited level of nitrogen is very close to the E5 level of CO2 atom, population in E5 level increases.

As soon as population inversion is reached, any of the spontaneously emitted photon will trigger laser action in the tube. There are two types of laser transition possible.

1.Transition E5 to E4 :

This will produce a laser beam of wavelength 10.6µm

2.Transition E5 to E3

This transition will produce a laser beam of wavelength 9.6µm. Normally 10.6µm transition is more intense than 9.6µm transition. The power output from this laser is 10kW.

Characteristics:

1. Type: It is a molecular gas laser.
2. Active medium: A mixture of CO2 , N2 and helium or water vapour is used as active medium
3. Pumping method: Electrical discharge method is used for Pumping action
4. Optical resonator: Two concave mirrors form a resonant cavity
5. Power output: The power output from this laser is about 10kW.
6. Nature of output: The nature of output may be continuous wave or pulsed wave.
7. Wavelength of output: The wavelength of output is 0.6µm and 10.6µm.

Advantages:

1. The construction of CO2 laser is simple
2. The output of this laser is continuous.
3. It has high efficiency
4. It has very high output power.
5. The output power can be increased by extending the length of the gas tube.

Disadvantages:

1. The contamination of oxygen by carbon monoxide will have some effect on laser action
2. The operating temperature plays an important role in determining the output power of laser.
3. The corrosion may occur at the reflecting plates.
4. Accidental exposure may damage our eyes, since it is invisible (infra red region) to our eyes.

Applications:

1. High power CO2 laser finds applications in material processing, welding, drilling, cutting soldering etc.
2. The low atmospheric attenuation (10.6μm makes CO2 laser suitable for open air communication.
3. It is used for remote sensing
4. It is used for treatment of liver and lung diseases.
5. It is mostly used in neuro surgery and general surgery.
6. It is used to perform microsurgery and bloodless operations.

5.7 Applications of Lasers

Lasers are widely used in the fields of

(1)Communication

(2)Computers

(3)Industry

(4)Scientific Research

(5)Military Operation

(6)Medicine

Lasers in Communication:

(1)In case of optical fiber communication, semiconductor laser diodes are used as optical sources and it's band width is very high compared to the radio and microwave communications.

(2)Laser is highly directional and has less divergence, hence it has greater potential use in space crafts and submarines.

(3)Lasers are also used in other communication devices, including high speed photo copiers and printers.

Lasers in Computers:

(1)In local area network (LAN), data can be transferred or transmitted from the memory storage of one computer to other computer using laser for short time.

(2)Lasers are used in CD-Rom's during recording and reading the data.

Lasers in industry:

(1)Lasers can be used to make holes in diamonds and hard steel.

(2)Lasers are used to cut teeth saws, drill eyes in surgical needles, guide bulldozers and test the quality of a fabric.

(3)They are used as a source of intense heat.

(4)Lasers range finder is used to measure distance to make map by surveyors.

(5)Lasers can cut, drill, weld, remove metal from surfaces and perform these operations even at surfaces inaccessible by mechanical methods.

Lasers in Scientific Research:

(1)Scientists are working on separating isotopes of uranium using laser.

(2)Lasers are used in the field of 3D-photography called holography.

(3)Lasers are used to produce certain chemical reactions.

(4)Using lasers, the internal structure of microorganisms and cells are studied very accurately.

(5)The laser has been used in Michelson-Morle experiment, which showed that the velocity of light was constant and thus showed the way for Einstein's theory of relativity.

Lasers in Military Applications:

(1)High energy lasers are used to destroy enemy air crafts and missiles.

(2)Lasers can serve as a war weapon.

(3)Laser can be used for detection and ranging like RADAR. The only difference is it uses light instead of radio waves. Hence it is called as Light Detecting and Ranging(LIDAR).

Lasers in medicine:

(1)Doctors use the heating action of a laser beam to remove diseased body tissue.

(2)Laser beam is used to correct a condition called retinal detachment by eye-specialist.

(3)Lasers are used in opening blocked arteries, reconnecting several nerves, removing warts and treating bleeding ulcers.

(4)Argon and CO_2 lasers are used in treatment of liver and lungs.

(5)Lasers are used for elimination of moles and tumors which are developing in the skin tissue.

Fiber Optics

Introduction:

The optical fiber is a cylindrical wave-guide system through which optical signals can be transmitted over long distances. It is playing an important role in the field of communication to transmit voice, television and digital data signals from one place to another. Fiber optics is important because they have

(1)High information carrying capacity.

(2)Light in weight and small in size.

(3)No possibility of internal noise and cross talk generation.

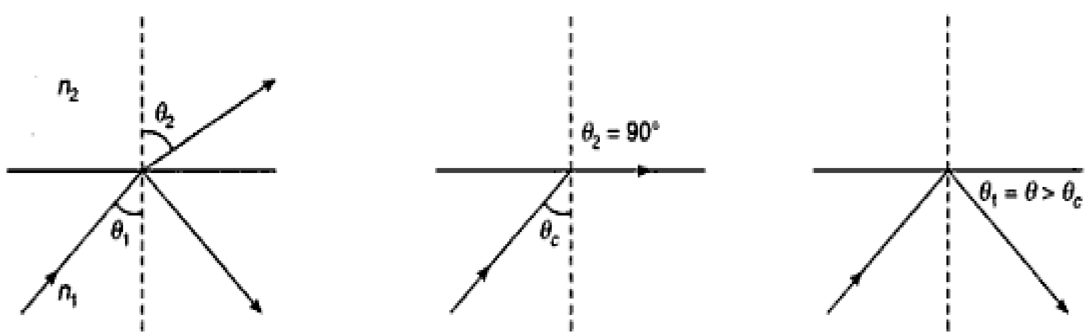
(4)No possibility of short circuits.

(5)Low cost of cables.

5.8 Principle of Optical Fiber

Total Internal Reflection:

The transmission of light in an optical fiber is based on the phenomenon of total internal reflection. According to total internal reflection, when a ray of light travels from a denser medium into a rarer medium and if the angle of incidence is greater than the critical angle then the light gets totally reflected. That is if the light ray is incident at an angle greater than the critical angle for the two media, the ray is totally reflected back into the medium by obeying the laws of reflection. This phenomenon is known as total internal reflection.



(a) Some part of the incident beam is transmitted through n_2, when $\theta_1 < \theta_c$ (b) Refracted ray passes through the core-cladding interface, when $\theta_1 = \theta_c$ (c) Total internal reflection occurs when $\theta_1 > \theta_c$

So according to law of reflection,

$$n_1 sin\theta_1 = n_2 sin\theta_2$$

Here, $\theta_1 = \theta_c$, $\theta_c = 90^0$

$$\therefore n_1 sin\theta_c = n_2 \sin 90^0$$

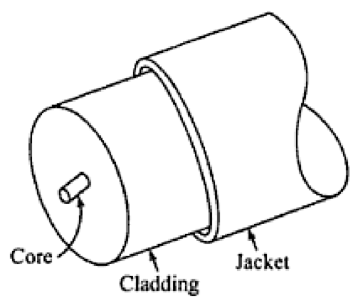
$$\Rightarrow \quad sin\theta_c = \frac{n_2}{n_1}$$

$$\Rightarrow \quad \theta_c = sin^{-1}\left(\frac{n_2}{n_1}\right) \text{ -------------- >(I)}$$

Here eq^n. (I) is the condition for total internal reflection.

5.9 Construction of Optical Fiber

Optical fiber is a very thin and flexible medium having a cylindrical shape consisting of three sections. They are



(1)The core material

(2)The cladding material

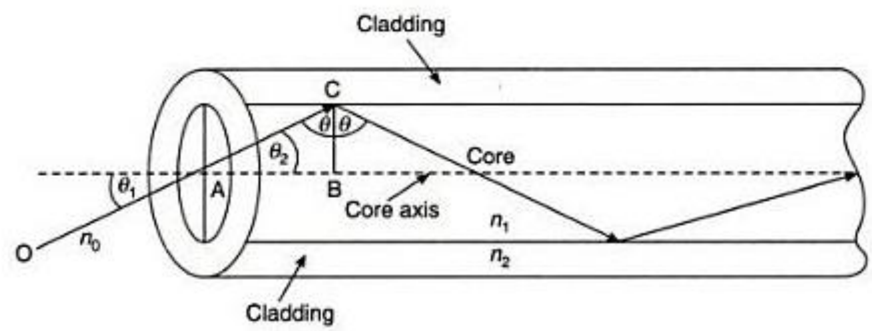
(3)The outer protective jacket

The fiber has a core surrounded by a cladding material. The refractive index is slightly less than that of core material, to satisfy the condition of total internal reflection. To protect the fiber material and also to give mechanical support there is a protective cover called as outer jacket.

5.10 Acceptance angle and Numerical Aperture

Acceptance angle:

Acceptance angle is defined as the angle at which we have to launch the light beam at one end of the fiber optic cable so that the entire light is propagated through the core.



Now in order to calculate an expression for the acceptance angle, consider the above figure which shows a cross section of a fiber with a light ray entering it. The light ray entered from a medium of refractive index n_o into core of refractive index n_1 .Here the ray (OA) enters with an angle of incidence α_i i.e,the incident ray makes an angle α_i with fiber axis.Let the refractive index of core be n_1 and the refractive index of cladding be n_2 .Here $n_1 > n_2$ and the light ray refracts at an angle α_r and strikes the core-cladding interface at angle θ .If the angle is greater than it's critical angle θ_c ,the light ray undergoes total internal reflection at the interface.

∴ According to Snell's Law, $n_o sin\alpha_i = n_1 sin\alpha_r$ ---------------- >(1)

From the right angled triangle ABC,

$$\alpha_r + \theta = 90^0$$

$$\Rightarrow \quad \alpha_r = 90^0 - \theta \quad \text{-------------- >(2)}$$

Substitute (2) in (1)

$$\Rightarrow \quad n_o sin\alpha_i = n_1 \sin(90^0 - \theta)$$

$$\Rightarrow \quad sin\alpha_i = \frac{n_1}{n_0} cos\theta \quad \text{-------------- >(3)}$$

When $\theta = \theta_c, \alpha_i = \alpha_m$ (maximum α value) i.e, maximum possible value of α_i for which $\theta = \theta_c$

$$\therefore \quad sin\alpha_m = \frac{n_1}{n_0} \cos\theta_c \quad \text{---------------- >(4)}$$

From eq^n of total internal reflection,

$$\Rightarrow \quad \sin\theta_c = \frac{n_2}{n_1}$$

$$\Rightarrow \quad \cos\theta_c = \sqrt{1 - sin^2\theta_c}$$

$$= \sqrt{1 - \left({}^{n_2}/_{n_1}\right)^2}$$

$$= \frac{\sqrt{{n_1}^2 - {n_2}^2}}{n_1} \text{ -------------------- >(5)}$$

From eq^{n}s. (4) & (5)

$$sin\alpha_m = \frac{\sqrt{{n_1}^2 - {n_2}^2}}{n_1} . \frac{n_1}{n_0} = \frac{\sqrt{{n_1}^2 - {n_2}^2}}{n_0}$$

For air $n_0 = 1$

$$\therefore \quad sin\alpha_m = \sqrt{{n_1}^2 - {n_2}^2}$$

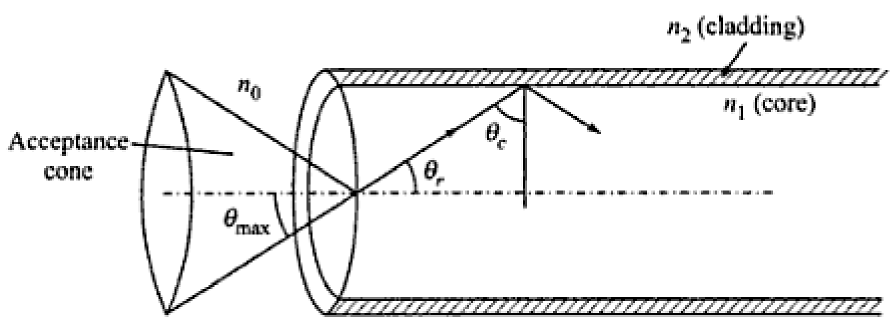
$$\Rightarrow \quad \alpha_m = sin^{-1}\sqrt{{n_1}^2 - {n_2}^2} \text{ --------------------- >(6)}$$

Here the angle α_m is called the acceptance angle.

Acceptance Cone:

It is the cone of light incident at the entrance end of the fiber which will be guided through the fiber, provided the semi vertical angle of the cone is less than or equal to α_m.

Diagrammatically it is shown in the below figure:



Numerical Aperture:

It is defined as the light gathering capacity of an optical fiber and is proportional to acceptance angle. Numerically it is equal to sine of acceptance angle.

Numerical Aperture, NA= $sin\alpha_m = \frac{\sqrt{{n_1}^2 - {n_2}^2}}{n_0}$

If the medium is air, then $n_0 = 1$

$$\therefore \quad NA = \sqrt{n_1^2 - n_2^2}$$

$$= \sqrt{(n_1 + n_2)(n_1 - n_2)}$$

$$= \sqrt{(n_1 + n_2) n_1 \Delta}$$

Where $\Delta = \frac{(n_1 - n_2)}{n_1}$ called as fractional difference in refractive indices n_1 and n_2

For most of the fibers $n_1 \approx n_2$

So we can take $n_1 + n_2 = 2n_1$

$$\therefore \quad NA = \sqrt{2n_1^2 \Delta} = n_1 \sqrt{2\Delta}$$

Therefore the numerical aperture is a measure of the amount of light that can be accepted by a fiber. It depends only on the refractive indices of core and cladding materials and is independent of fiber dimensions and generally this value ranges from 0.1 to 0.5.

5.11 Classification of fibers:

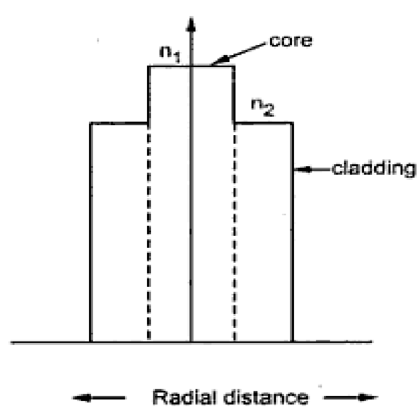
Depending upon the refractive index profile of the core, optical fibers are classified into two categories, they are

(1)Step-index fibers

(2)Graded-index fibers

Step-index fibers:

In step-index fibers, the refractive index of the core medium is uniform through out and undergoes a sudden change at the interface of core and cladding medium.



There are two types of step-index fibers, they are

(1)Single-mode step index fibers

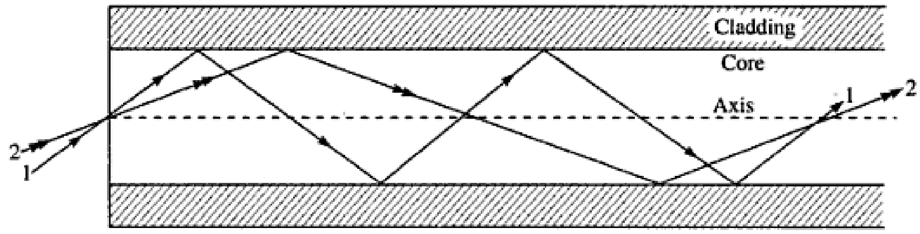
(2)Multi-mode step index fibers

The diameter of the core for a single mode step index fiber is very small of the order

10µm and it propagates the light in only one mode, whereas the diameter of the core for a multimode fiber is of the order 50-200 μm , and it propagates the light in multiple modes.

Transmission of signal in step-index fibers:

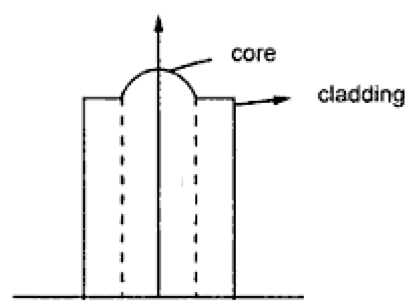
In digital communication systems, information is transmitted in the form of pulses. Light pulses transmitted through the fiber are to be decoded at the receiving end to receive the information.



Two rays entering a step-index fibre at different angles.

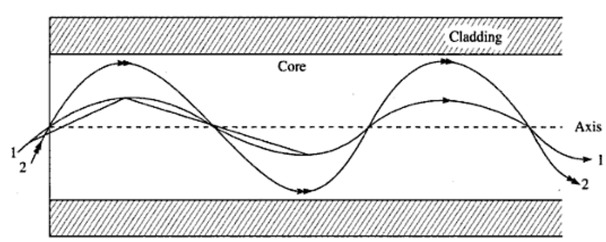
As shown in the above figure, there are three such pulses travelling through the fiber striking the core-cladding interface at different angles and they travel along the fiber by multiple reflections. They have different paths and reach the other end of the fiber at different points i.e., the three rays reach the received end at different times. The pulsed signal received at the other end will be broadened and this is called intermodal dispersion. This reduces the transmission rate and capacity. So smaller the pulse dispersion greater will be the information transmission capacity of the system.

Graded-index fibers: In graded-index fibers, the refractive index of the core medium is made to vary in a parabolic manner such that the maximum refractive index is present at the centre of the core. The diameter of the core is about 50 μm , therefore only multimode propagation is possible in graded index fibers.



Transmission of signal in Graded-index fiber:

In graded index fibers, optical signals travel through core medium in the form of skew rays or helical rays. As shown in the below figure, though different rays travel through the fiber but all come to focus at the same point i.e., all the rays arrive at the other end of the fiber at the same time. This is possible because the velocity is inversely proportional to refractive index.



Two rays entering a GRIN fibre at different angles.

5.12 Attenuation in Optical fibers

Different mechanisms are responsible for the signal attenuation of the fiber and these mechanisms are influenced by material compositions, purification level, wave guide structure.

The attenuation of signal is measured in decibel/km. Signal attenuation is defined as the ratio of the input optical power P_i into the fiber to the output received optical power P_o from the fiber.then the attenuation coefficient of the signal per unit length is given as

$$\frac{\alpha = \frac{10}{L}\log P_i}{P_o} \frac{db}{km}$$

Where $L \rightarrow$ length of fiber

The mechanism of attenuation of signal may be classified broadly into two categories, they are

(1)Absorption losses

(2)Scattering losses

Absorption losses:

It is a material property i.e., it is a characteristic possessed by all the materials. Every material in the universe absorbs few suitable wavelengths as they incident on the material or passed through the material. In the same way core material of a fiber absorbs few wavelengths as the optical pulses or wavelengths pass though it.

Scattering losses:

The core medium of fibers is made of glass or Silica crystalline materials. Since there is no ideal crystal in the Universe, this medium possess few crystal defects. So in the passage of optical signals in the core medium if crystal defects are encountered, they deviate from the path and the total internal reflection is discontinued. Hence such signals will be destroyed by entering into the cladding.

Bending Losses:

The distortion of the fiber from the ideal straight line configuration may also result in losses in fibers. Tight bands cause some of the light to not to be internally reflected but to propagate into the cladding and be lost. The optical power scattered out of fiber at a major bends depends exponentially on the bend radius(R).The loss coefficient can be represented as

$$\alpha_B = C\ exp\left(-\frac{R}{R_C}\right)$$

Where 'C' is a constant and $R_C = \frac{a}{(NA)^2}$

Where 'a' is the radius of fiber.

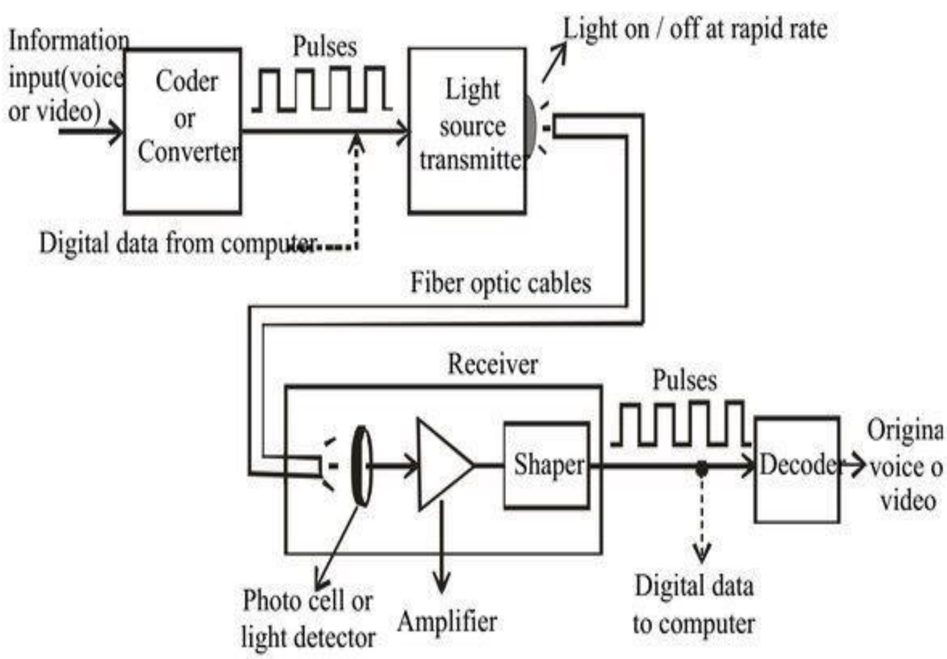
Micro bending and Waveguide losses:

In fiber optical cables a continuous succession of very small bends may cause a very significant rise in fiber attenuation. This effect is known as micro bending loss. Continuous small variation in core diameter or circularity which can easily arise during manufacture if sufficient care is not taken, gives rise to a similar scattering mechanism and cases losses which are called as wave guide losses.

By taking proper precaution all these losses can be minimized and they are largely wavelength independent.

5.13 Optical Fiber Communications

The communication system of fiber optics is well understood by studying the parts and sections of it. The major elements of an optical fiber communication system are shown in the following figure.



Itessentially consists of following parts.

1.Encoder 2.Transmitter 3.Waveguide 4.Receiver 5.Decoder

Encoder: It converts electric signal corresponding to analog information such as voice, figures, objects etc into a binary data This binary data comes out in the form of stream of electrical pulses.

Transmitter: It mainly consists of drive circuit and a light source Drive circuit supplies the

electric pulses to the light source from the encoder LED or diode laser is used as light source and it converts electrical signals are infected into optical signals These optical signals are injected into wave guide.

Wave Guide: It carries the information through the desired distance in the form of optical signal.

Receiver: It consists of photo detector, amplifier, signal restorer

Photo Detector: It converts optical signal into electrical signal This signal may become weak since it travels through very long distance

Amplifier: Such weak signal from photo detector is amplified This is allowed into signal restorer

Signal Restorer: The function of signal restorer is to put the signals In order which are received from wave guide subsequently from photo detector

Decoder: Finally, signals will be decoded and sent in the original form

5.14 Applications of Optical Fiber

Fiber optic cables find many uses in a wide variety of industries and applications. Some uses of fiber optic cables include:

- **Medical**

Used as light guides, imaging tools and also as lasers for surgeries

- **Defense/Government**

Used as hydrophones for seismic waves and SONAR , as wiring in aircraft, submarines and other vehicles and also for field networking

- **Data Storage**

Used for data transmission

- **Telecommunications**

Fiber is laid and used for transmitting and receiving purposes

- **Networking**

Used to connect users and servers in a variety of network settings and help increase the speed and accuracy of data transmission

- **Industrial/Commercial**

Used for imaging in hard to reach areas, as wiring where EMI is an issue, as sensory devices to make temperature, pressure and other measurements, and as wiring in automobiles and in industrial settings

- Broadcast/CATV

Broadcast/cable companies are using fiber optic cables for wiring CATV, HDTV, internet, video on-demand and other applications

Short Questions

1. What is LASER? Explain the characteristics of LASER's.
2. Explain the phenomenon of Absorption, Spontaneous emission and stimulated emission.
3. Explain different pumping mechanisms in laser.
4. Explain working principle of laser.
5. What are the main components of laser.
6. What is population and population inversion?
7. What is optical fiber? Explain the principle of optical fibres.
8. Explain construction of optical fibre.
9. What are step index and graded index optical fibres?
10. What are single mode and multi mode optical fibres?
11. Explain the Attenuation in Optical Fibers
12. What are radiative and nonradiative recombinationmechanism?
13. Write short note on solar cell.
14. Explain construction and working of LED.
15. Write the applications of solar cell

Essay type Questions

1. What are Einstein's coefficients of radiation? Give Einstein relation of radiation.
2. Describe the construction and working principle of ruby laser.
3. Describe the construction and working principle of He-Ne laser.
4. Describe the construction and working principle of CO2 laser.
5. Explain in detail acceptance angle, acceptance cone and numerical aperture of optical fibre.
6. Classify and explain different types of optical fibres with neat light ray diagrams.
7. Explain the applications of optical fibres.
8. Explain construction and working of semiconductor laser.
9. Explain PIN diode in detail.
10. Discuss avalanche photo diode in detail.

Multiple answer questions

1. The wavelength emitted by He-Ne laser is [ ]

(A) 623.8 nm (B) 632.8 nm (C) 682.3 nm (D) 683.2 nm

2. If N_1 and N_2 are the population densities of lower energy state E_1 and higher energy state E_2 respectively, the laser emission is possible only when [ ]

(A) $N_2 > N_1$ (B) $N_2 < N_1$ (C) $N_2 = N_1$ (D) no relation between N_1 and N_2

3. Brewster law is [ ]

A) $\mu = \tan\theta_p$ C) $\mu > \tan\theta_p$
B) $\mu < \tan\theta_p$ D) None of these

4. The active element in _________________ laser is chromium. [ ]

A) ruby B) He-Ne C) Co2 D) Semi conductor

5. Propagation of light through fiber core is because of ________________. [ ]

A) Reflection B) refraction C) Total internal reflection D) None

6. The condition for lasing action is [ ]

(A) Excitation (B) meta-stable state (C) population inversion (D) emission

7. In the absence of degeneracy, Einstein coefficients B_{12} and B_{21} are related as [ ]

(A) $B_{12} / B_{21} = h$ (B) $B_{12} = B_{21}$

(C) $B_{12} / B_{21} =$ constant (D) $B_{21} / B_{12} = h$

8. The refractive index of core is 1.5. The refractive index of cladding can be [ ]
(A) 1.52 (B) 1.55 (C) 1.60 (D) 1.45

9. In an optical fibre, meridinal rays propagate [ ]
(A) along the walls of the fiber (B) along the axis of the fiber

(C) into the cladding material (D) along the cladding of the fiber

10. The active medium in He-Ne laser is _______________ [ ]
A) Ne B) Cr C) GaAs D) None of these
11. In semiconductor laser, the population inversion is achieved by heavily doped [ ]
A) Ne B) Cr C) GaAs D) None of these
12. The acceptance angle of an optical fibre having numerical aperture 0.5, is_ [ ]
A) 30^0 B) 60^0 C) 90^0 D) None
13. The ratio of Einstein's coefficient A_{21}/ B_{21} [ ]

a) $\frac{8\pi h v3}{c3}$ b) $\frac{8\pi h v3}{c2}$ c) $\frac{8\pi h v3}{c}$ d) $\frac{2\pi h v3}{c}$

14.The source of excitation in ruby laser is [ ]

a) Sodium vapor lamp b) Xenon flash lamp c) Mercury vapor lamp

d) Incandescent lamp

15.The attenuation in an optical fiber is a function of [ ]

a) Fiber material only b) Wavelength of light only

c) Length of the fiber only d) All the above

16. __________ Laser is used in optical communications [ ]

A) Ruby B) He-Ne C) Co2 D) Semi conductor

17. The population of the various energy levels of a system in thermal equilibrium is given by [ ]

(a) Boltzmann distribution law (b) Einstein relations

(c) Planck's law (d) Beer's law

18 . T.Maiman invented [ ]

(a) He-Ne laser (b) CO2 laser

(c) Ruby laser (d) Nd: YAG laser

19. The colour of the laser output from a Ruby laser is [ ]

(a) green (b) blue (c) red (d) violet

20.. In He-Ne lasers, the ratio of He-Ne is in the order [ ]

(a) 1:10 (b) 1:1 (c) 100:1 (d) 10:1

21 Laser radiation is [ ]

(a) monochromatic (b) highly directional

(c) coherent and stimulated (d) All

22. Propagation of light through fiber core is due to [ ]

(a) diffraction (b) interference (c) refraction

(d) total internal reflection

23. Step index fiber can be a [ ]

(a) monomode fiber only b) multimode fiber only

(c) monomode as well as multimode fiber

(d) either monomode or multimode

24. The process of mixing the signal with the carrier is called [ ]

(a) dispersion (b) attenuation (c) modulation

(d) demodulation

25. To overcome the problem of inter model dispersion one has to use [ ]

(a) multimode fiber (b) step index fiber

(c) graded index fiber (d) plastic fiber